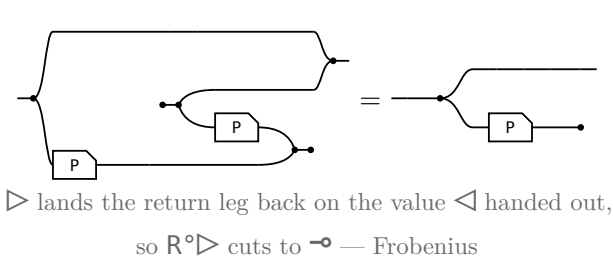


Relation Algebra

§5. Domain and range

The **domain** $\text{dom}(R) \triangleq \mathbb{1} \cap R R^\circ$ and the **range** $\text{ran}(R) \triangleq \text{dom}(R^\circ)$.

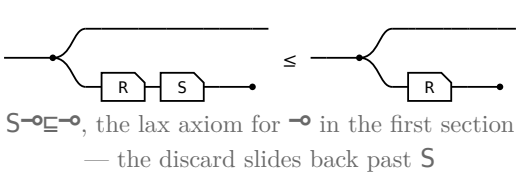


Running R and throwing the result away leaves only the fact that R could fire, and $\text{ran}(R)$ is the same picture with the box mirrored. In Rel both steps are $\{(a, a) : \exists b. a R b\}$.

$\text{dom}(R) \triangleq \mathbb{1} \cap R R^\circ$
$\text{Coreflexive}(X) \implies \text{dom}(R) \sqsubseteq X \iff R \sqsubseteq X R$
$\text{dom}(RS) \sqsubseteq \text{dom}(R)$
$\text{dom}(R \cap S) = \mathbb{1} \cap S R^\circ$
$R \text{ entire} \iff \text{Entire}(R) \triangleq \text{dom}(R) = \mathbb{1}$
$R \text{ simple} \iff \text{Simple}(R) \triangleq R^\circ R \sqsubseteq \mathbb{1}$
$R \text{ a map} \iff \text{Map}(R) \triangleq \text{Entire}(R) \wedge \text{Simple}(R)$
$\text{Entire}(R) \wedge \text{Entire}(S) \implies \text{Entire}(RS)$ $\text{Simple}(f) \wedge \text{Simple}(g) \implies \text{Simple}(fg)$ $\text{Map}(f) \wedge \text{Map}(g) \implies \text{Map}(fg)$
$\text{Entire}(RS) \implies \text{Entire}(R)$

§5.1. Sliding the discard

$\text{dom}(RS) \sqsubseteq \text{dom}(R)$, and a single glyph for **dom** would have nothing to slide: with the box and the discard drawn apart, the law is one dot walking back along the lower strand.



Equality is S entire, which is the same picture read as $\text{dom}(R) = \mathbb{1} \iff R \text{ entire}$.